Shaker Tarneem Refaat Fawzi

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www.linkedin.com/in/tarneem-shaker-33448220a | Portfolio: Tarneem Shaker — Mechanical Engineer

Mechanical Engineering graduate with hands-on experience supporting maintenance and quality operations in automotive manufacturing. Skilled in SolidWorks, MATLAB, and G-code, with practical knowledge of preventive maintenance, fault diagnosis, mechanical design, and 8D problem-solving. Interested in applying engineering skills to improve equipment reliability, manufacturing efficiency, and production processes.

EXPERIENCE

Maintenance and Production Trainee | Teriak Group
June 2025 - September 2025

- Assisted maintenance technicians with preventive maintenance and fault diagnosis on automotive parts manufacturing equipment across production lines
- Performed in-process quality inspections to verify components met IATF 16949 quality-management requirements, flagging deviations to senior staff
- Observed and assisted with root-cause troubleshooting on recurring equipment issues alongside senior maintenance engineers

EDUCATION

Bachelor of Mechanical Engineering | University of Dunaújváros

- June 2026 | GPA: 4.52/5.0

Engineering Prep Program | Budapest International College

- June 2023

American High-School Diploma | Nile Union Academy

- June 2021 | GPA: 3.55/4.00

PROJECTS – For more projects and details: https://tarneemshaker.github.io/tarneem-portfolio/projects.html

G-Code Toolpath Generator & Analyzer | Personal Project

- Built a Python command-line tool that generates and analyzes CNC G-code for pocket milling operations, applying toolpath logic (multi-pass depth cutting, tool radius compensation) and Python fundamentals (argument parsing, regex-based file parsing, geometry calculations)

Footstep Energy Harvesting System | BSc Thesis Project

- Designed the mechanical assembly of a rack-and-pinion footstep energy harvesting system in SolidWorks, including the loading platform, spring return mechanism, transmission housing, flywheel, and structural frame.
- Performed fatigue life analysis of critical components (springs, gears, shafts, bearings) using S-N curve analysis and Miner's cumulative damage rule applied to a realistic pedestrian load spectrum; identified the spring as the system-limiting component
- Conducted FMEA to rank failure modes by risk priority and developed a structured preventive maintenance plan; proposed design improvements including material upgrades, shot peening, and sealed bearing specification to extend service life

SKILLS

- CAD & Design: SolidWorks, Siemens NX, Mechanical Assemblies
- G-Code / CNC fundamentals
- 8D Analysis - root cause & corrective action
- Python (scripting, CNC/G-code automation)
- ERP: SAP PP / SAP PM – Fundamentals
- MATLAB – Numerical Computing & Engineering Analysis

LANGUAGES

- **Arabic** – Native | **English** – Professional | **French** – Intermediate | **Hungarian** – Beginner